Ultrasound (US) – Basic Concepts

Definition of Ultrasound

Ultrasound is:

☞ **Mechanical sound waves with frequencies higher than the upper limit of human hearing**

- Human hearing range: **20 Hz → 20 kHz**
- Ultrasound: **> 20 kHz**

☞ In medical applications:

- Typically used frequency: **1 MHz to 15 MHz**

Medical Applications of Ultrasound

1. **Obstetrics (Pregnancy)**
   ☞ Monitoring the fetus
2. **Cardiology**
   ☞ **Echocardiography**
3. **Abdominal Imaging**
   ☞ Liver
   ☞ Kidneys
4. **Blood Vessels**
   ☞ **Doppler Ultrasound**

Doppler Ultrasound

Used to:

☞ Measure **blood velocity**

Doppler Equation

$$f_D = \frac{2 f_0 V \cos\theta}{c}$$

Where:

- **f_D** = Doppler frequency shift
- **f_0** = transmitted frequency
- **v** = blood velocity
- **θ** = angle between beam and flow
- **c** = speed of sound

Advantages of Ultrasound

1- Safe (non-ionizing)
2- Fast
3- Inexpensive
4- Portable
5- Real-time imaging

Limitations of Ultrasound

A- Cannot penetrate bone
B- Cannot pass through air

Physical Nature of Ultrasound

Ultrasound waves are:

☞ **Mechanical longitudinal waves**

This means:

- They require a **medium** to propagate
- They **cannot travel in vacuum**

They propagate through:

A Soft tissues
A- Fluids
B- Organs

Components of an Ultrasound System

1. **Transducer (Probe)**
   ☞ Acts as both **source and detector**
   ☞ Contains a **piezoelectric crystal**
2. **Pulser**
   ☞ Sends short electrical pulses
3. **Receiver**
   ☞ Receives reflected signals
4. **Signal Processor**
   ☞ Processes the signal
5. **Monitor**
   ☞ Displays the image

Function of the Pulser

The pulser:

☞ Sends short electrical pulses to the crystal

When the pulse reaches the crystal:

☞ The crystal vibrates
☞ Produces ultrasound waves

This happens **before entering the body**

What Happens Next?

After emission:

- The pulser stops
- The wave enters the body

Then:

① The wave propagates through tissues
② It encounters tissue boundaries
③ Part of it is reflected
④ The echo returns to the crystal

When the Wave Returns

The crystal now acts as a:

☞ **Detector**

- The returning wave compresses the crystal
- The crystal generates an electrical signal
- The signal is sent to the receiver

Correct Sequence of Operation

① Pulser sends electrical pulse
② Crystal generates ultrasound wave
③ Wave enters the body
④ Wave reflects back
⑤ Crystal receives the signal
⑥ Receiver processes the signal

Important Note

The same crystal functions as:

Source (during transmission)
Detector (during reception)

☞ Therefore, it is called a **Transducer**

Piezoelectric Effect

Inside the crystal:

- Applying voltage → crystal vibrates → produces ultrasound
  ☞ **Piezoelectric effect**

When echoes return:

- Pressure on crystal → generates voltage
  ☞ **Inverse piezoelectric effect**

How Ultrasound Image is Formed

① The transducer emits sound waves
② Waves enter the body
③ They interact with tissues
④ Reflections occur
⑤ Echoes return to the probe
⑥ Converted into electrical signals
⑦ Processed by the computer
⑧ Image is displayed

Why Do Waves Reflect?

Due to:

☞ Differences in **acoustic impedance (density differences)**

Example:

- Bone
- Muscle
- Fluid

☞ Each reflects sound differently

Important Equation in Ultrasound

Wave Equation

V= f λ

Where:

- **v** = (wave speed)
- **f** = (frequency)
- **λ** = (wavelength)

Speed of Ultrasound in the Body

☞ Approximately:

1540 m/s

☞ This value is crucial for **depth calculation**

Depth Calculation

$d = \frac{vt}{2}$ Where:

- **d** = depth
- **v** = speed of sound
- **t** = time

☞ We divide by 2 because the wave:

- Travels to the tissue
- Then returns back

Example

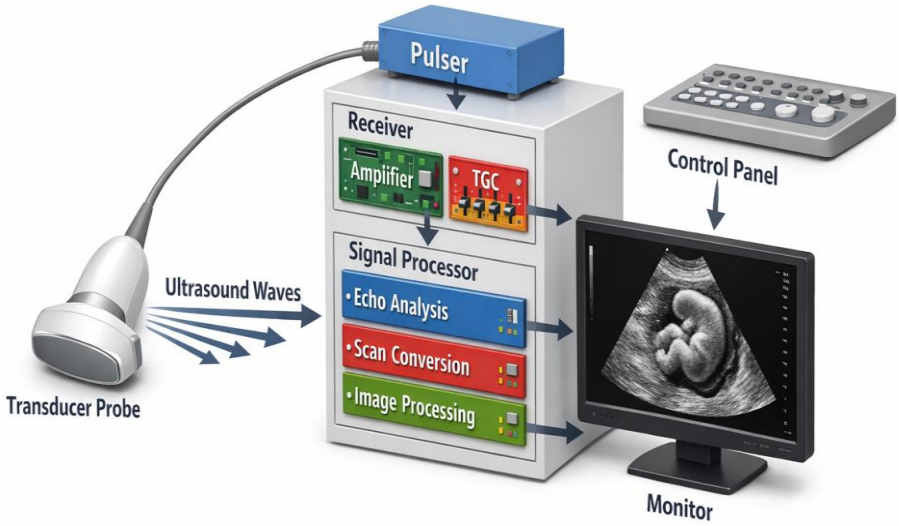
If:

- **t = 200 µs**
- **v = 1540 m/s**

Then:

$$d = \frac{1540 \times 200 \times 10^{-6}}{2}$$

$$d = \mathbf{0.154\ m} = \mathbf{15.4\ cm}$$



Ultrasound Imaging System Components (3D View)